Digital platform to manage and engage with alumni in a centralized manner

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Abstract- Keeping contact with the alumni and managing the information has been a major challenge in the digital age to the educational institutions. Conventional methods of managing alumni are mostly based on their scattered databases, manual files, and informal channels of communication making the information available to them out of date, and with less interaction between the institutions and their alumni. In this paper, the authors suggest a Digital Platform of Centralized Alumni Data Management and Engagement aimed at simplifying the storage, communication, and collaboration of alumni data in the same platform.

To make the proposed platform scalable, secure, and maintainable, a multi-layered architecture has been used and it comprises of presentation layer, application layer, data layer and analytics engine. The system incorporates some of the most important modules such as alumni data management, alumni-student interaction, job referral services, and administrative monitoring which is allowing mentorship opportunities, professional networking, and effective monitoring of the alumni. Moreover, the platform includes analytics dashboards to show insights into the alumni engagement patterns and the effectiveness of the institutional outreach.

The adoption of this centralized system enhances the accuracy of the data, boosts the turnout of the alumni, and aids institutional development based on the data. The proposed solution illustrates that the digital transformation can enhance the relationship with the alumni, bring better opportunities of networking in the career of students and help in the long-term development of an institution.

Keywords-- Alumni Management System, Digital Alumni Platform, Alumni Engagement, Centralized Data Management, Alumni-Student Interaction, Job Referral System, Data Analytics.

I. INTRODUCTION

One of the most useful strategic resources of educational institutions is alumni networks. Graduates also play an important role in the development of the institution by getting involved in mentorship, professional networking, knowledge sharing, financial support, and collaboration in the industry. Good relationships between alumni may boost the reputation of the institution, better placement of the students and build academic-industry relationships. Nonetheless, the issue of keeping and sustaining the alumni contact with large and widely dispersed geographies is also a significant challenge to most institutions.

Conventional management practices of alumni are not that innovative and are usually based on piecemeal databases, spreadsheets, social media groups, and manual means of communication. Such practices often lead to obsolete documentation, lack of unity in communication, and the lack of decent chances to engage the alumni meaningfully. A number of surveys point out that lack of centralized systems does not allow institutions to efficiently track alumni career building, participation and contribution towards institutional development [3], [5]. Moreover, there is a high risk of data redundancy, inaccuracies, as well as inefficient information retrieval when managing records manually.

The recent studies underline the great significance of digital platforms in alumni management and engagement. It has been demonstrated that interactive alumni tracking dashboards can help institutions to track the outcomes of alumni career development and determine the success of academic programs [1]. Likewise, the contemporary alumni web sites incorporate networking capabilities, communication portals and profile management systems to enable alumnus and present day students to collaborate with each other [8], [9]. Alumni engagement has been further increased with the introduction of smart technologies including artificial intelligence and data analytics that allow creating personal communication and

automated recommendations [2], [7].

There are also web-based alumni management systems, which have been suggested to allow the graduates to update their profile, communicate about their professional success and take part in the activities of the institution by means of centralized repositories [3], [4]. Such systems enhance the availability of data and the institutions are able to keep orderly alumni records. In addition, with the help of sophisticated analytics, administrators can study the trends of the engagement of alumni, assess the effectiveness of institutional activity, and help make strategic decisions [6], [10].

Nevertheless, in spite of such developments, some of these current alumni systems are yet to be fully integrated with centralized data management, modules of engagement, analytic capability, and monitoring by administrators into a single architecture. In a bid to fill this gap, this study presents a Digital Platform of Centralized Alumni Data Management and Engagement that will simplify management of alumni information and enhance cooperation among the alumni, students and institutional administrators. The suggested system incorporates the multi-layered design answering the scalability requirements, security, and effective performance of the system in addition to the alumni-student communication, mentorship possibilities, job referral services, and analytics-driven insights.

The proposed platform is intended to build stronger relationships among alumni, improve career outcomes among students, and contribute to the long-term development of the institution by providing data-driven alumni engagement approaches.

II. LITERATURE REVIEW

The alumni management systems have been of great interest in the recent years owing to the growing relevance of the alumni networks in providing support of institutional growth, mentoring and career development opportunities. Educational institutions are realizing that alumni can be a key strategic stakeholder in terms of knowledge sharing, placement and the reputation of the institution. Consequently, some of the studies have covered how digital platforms can be developed in managing and engaging alumni data.

Sanusi et al. [1] suggested an interactive dashboard of alumni tracking that would track the progress in career development and institutional influence of the alumni. Their work further highlighted the significance of visualization tools that can assist institutions to analyze the results of alumni and make viable strategic decisions. It was proved that centralized dashboards enhance the accessibility of data and lead to improved monitoring of the trends of alumni engagement.

Loganathan et al. [2] proposed an AI-powered alumni portal, which adds intelligent recommendation systems to facilitate networking and professional interaction between alumni. The platform facilitates automated interaction, customized recommendations, and professional networking, which enhance the involvement of the alumni and interaction in the institutional ecosystem.

Bista et al. [3] designed an alumni tracking system web based enabling the alumni to update their professional details and actively retain a relationship with their college. The system offers a central repository of alumni information which aids institutions to have proper records of this information and minimizes redundancy of data.

In the same vein, Babu et al. [4] were interested in the design of an alumni portal that would be more secure in terms of data security. Their study focused on the need of safe authentication, encoded communication and role-based access control as a guarantee of safety of confidential alumni data in the digital environment.

Luciano et al. [5] developed a tracking alumni system at the level of higher education institutions, which employs the collection of alumni data, the tracking of their careers and communication. This paper has shown how information systems technology can be used to streamline administrative operations of the alumni and enhance institutional outreach.

Trillano et al. [6] came up with a system of alumni tracer and management system with data analytics, which allow institutions to track patterns of graduate employability and assess the efficacy of academic programs. Their study emphasized the potential of analytics in assisting evidence-based decision-making in the institutions.

Yumen [7] examined how a regression model and predictive analytics can be applied to alumni network platforms to analyze alumni data and determine trends of their engagement. The research proposed that sophisticated analytics can help institutions predict the alumni engagement and enhance the selective communication approach.

Vishwakarma et al. [8] presented a platform of alumni association that was meant to increase communication among the alumni and institutions using online portals. The system provides event management, channels of communication and networking with graduates.

Maya et al. [9] came up with an Alumni Connect Portal, which seeks to build on the effectiveness of the Alumni activities by offering an interactive functional platform to enhance the engagement of the alumni in terms of profile management, discussion-forums, and institutional updates. Their system shows that digital platforms can create better alumni communities.

Deepika and Santhi [10], examined how alumni engagement platform influences career development, and they found out the potential of alumni networks in offering mentorship, internships, and exposure to the industry among the students. According to their study, structured alumni engagement systems can significantly enhance the employability of students and the potential to get access to professional networking.

Though they offer some important findings as to the technologies of alumni management, a significant portion of the existing solutions is dedicated to separate aspects like the alumni tracking, networking, or analytics. It is still necessary to have a platform with a modular and end-to-end alumni management, engagement, analytics, and administrative overview of the system, which has scalability and is secure. In this study, the researcher fills this gap by developing a Digital Platform of Centralized Alumni Data Management and Engagement that incorporates all these features in a single platform that is aimed at improving alumni relations and assisting in the growth of the institution.

III. METHODOLOGY

The Online System of consolidated Alumni Data Management and Engagement is developed according to a structured workflow as reflected in figure 1 because it requires reliability, scalability, and security. The methodology has been further sub-divided into a few phases the first one being the requirement analysis and the last one being the system deployment and evaluation. The phases to be incorporated in developing a sound system will include each phase that will effectively manage the alumni information and enhance the communication between the alumni, students and the administrators.

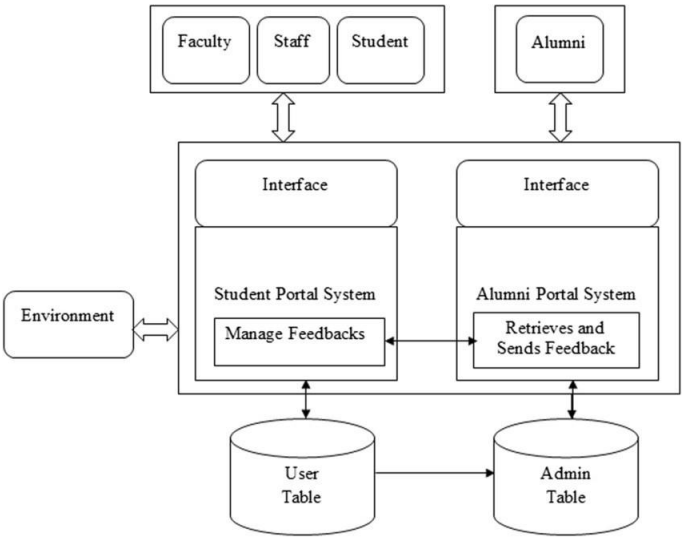

Figure 1: Proposed System Methodology Workflow

A. Requirement Analysis

Identification of non-functional and functional system requirements is carried out in the first phase. The surveys will be used to consult the stakeholders which will include the alumni, students, placement officers and the administrative staff as well as the workflow analysis. The main issues that may have been pointed out are poor alumni records, old contacts data and inability to track and poor analyties. The functional requirements are the management of alumni profile, attending event, mentorship communication, job referral service, and administration reporting. Another example of non- functional requirements is the system scalability, usability, security, performance efficiency and data privacy requirements.

B. Systems Architecture Design.

A multi-layered system architecture is developed using the obtained requirements in order to make the system architecture modular and maintainable. The structure is composed of the following building:

Presentation Layer - platform that offers both web and mobile access to the alumni, students and administrators.

Application Layer - It is concerned with business logic, API interaction, authentication and workflow control.

Data Layer - This is where a central relational database with a record of the data on the alumni and institutional data is stored.

Analytics Engine - Students Data students data is processed to provide insights, i.e., engagement patterns and career progression analysis.

This structure will be hierarchical which will make upgrades to the structure easier, better performance and fault isolation is attained.

C. Database Design

The demographic data, the academic data, the professional data, the event attendance and contribution data of the alumni will be stored in normalized database design. An attempt is made to make the data retrieval faster and ensure the validity of the data by using the indexing methods and the criteria of validation. With the acquisition of the centralized database, the duplication of the information is removed and the unique source of information about the alumni is offered.

D. Module Development and Integration

The system has a modular programming to make sure it is flexible and scalable. Major functional modules are:

1. Alumni Data management Module.
2. Intensive communication between the students and the alumni.
3. Job Opportunities and Referrals Module.
4. Administration Monitoring and Reporting Module.

These modules are also connected by the use of RESTful API and this is what allows the optimal connectivity between the front end interfaces and the services that are provided on the back end.

E. Security Implementation

Security is another issue that is very important about the platform. The system has several security controls that are inclusive of:

- Role-based access control (RBAC)
- Authenticated and nomadic passwords.
- Use of encryptions in communicating with HTTPS/SSL.
- Secure session management
- A lot of vulnerability testing.

These will guarantee the data security of the sensitive data of the alumni and will meet the data privacy standards

F. Testing and Validation

The system is verified at a number of levels in order to determine the effectiveness and efficiency of the system:

- **Unit Testing** - tests modules and parts.
- **Integration Testing** - The inter-module communication is provided by the integration testing.
- **System Testing** - Tests the system functionality.
- **User Acceptance Testing (UAT)** - Assures that the platform is required to meet the user requirements.

Tests on performance measures are response time, multiuser simultaneous and efficiency of database queries.

G. Deployment and Evaluation

Simultaneously when a successful test is achieved, the system is deployed on a safe server set up. The post deployment evaluation is to confirm that the systems are functioning according to the expectations and the users will give feedbacks and give a recommendation on the improvement of the systems. The reliability of the system will be guaranteed by constant monitoring which will make the system sustainable in the long- term.

IV. IMPLEMENTATION AND EVALUATION

A. System Implementation

The proposed Digital Platform of Centralized Alumni Data Management and Engagement was implemented following a modular architecture in order to be scalable, secure and maintainable. The system will integrate different components that include user interfaces, backend services and central database, to successfully carry out the functioning of the alumni information and engagements.

The contemporary web technologies were used in the presentation layer development to provide the alumni, students and the administrators with the user interface that is easy to use and respond to. The activities that can be performed through this interface include creation of profile, attending events, job referrals as well as mentorship communication.

It contains the core processing unit of the system which is the application layer and deals with the business logic. RESTful APIs are used to open and close the interaction between the frontend and the services on the backend. Application layer would be the one that would receive a user request, check the validity of the data and send it to the centralized database to either retrieve or update the information.

The data layer contains the past records of the alumni in the form of school history, work history, attendance history and involvement history. Normalization and the storage of the data in a relational database structure ensures the data integrity and elimination of redundancy.

The system consists of a number of functional modules:

Alumni Data Management Module - This will give the alumni an opportunity to update on personal and professional information.

Alumni-Student Interaction Module - Enables mentoring and communicating of the alumni and students.

Job Referral Module - This module allows sharing of job opportunities and career guidance by the alumni.

Administrative Monitoring Module - The module will provide the dashboards to allow the administrators to handle the alumni activities and generate reports.

Role-based access control (RBAC), passwords, and the application of the HTTPS communication protocols were also implemented to promote security of the information provided by the user and to offer security access to the system.

B. System Evaluation

The system was piloted to identify functionality, usability and system performance in regard to alumni information and engagement activity. The evaluation stage involved a lot of testing strategies.

The functionality test was conducted to ensure that all the modules; alumni registration, profile management, mentorship communication, job referral services and administrative monitoring, were functioning correctly.

Performance testing was carried out to estimate the response time of the system, efficiency in database query and the ability to handle a certain number of concurrent users. The database was very centralized and this increased the speed of data retrieval as well as reduced redundancy.

The Usability Testing was conducted by providing the opportunity of using the platform to the participants who were alumni, students, and administrators and comment on the user interface and system functionality. The results showed that the system provided a convenient platform to use in managing the alumni information and contact.

Security Testing was done to provide verification, access control rules and secure data transfer. Role-based access and encrypted communication was introduced to ensure the sensitive data of alumni was not leaked.

The outcomes of the analysis indicate that the suggested platform has managed to achieve better alumni data management, communication between the alumni and institutions, and make decisions based on the data gathered via centralized analytics. It can also be expanded to other capabilities such as mobile apps, AI-driven recommendations, and predictive analytics will be implemented into the system in the future.

V. RESULTS AND DISCUSSION

A new web-based system was proposed The Digital Platform, Unified Alumni Data Management and Engagement which was experimented to test the potential application in managing the alumni data and the likelihood of realising the connection between the students and the alumni. The system formulated has some of the modules, including user authentication, student dashboard, mentorship interaction and job referral services. The utility and the ease of the offered platform can be evidenced with the help of the results which were downloaded with the assistance of installed system.

A. User authentication and access control.

The authentication interface is the point of entry of the users who consist of the students, alumni and the administrators. On the platform, the system steals the user credentials that include user ID, email address and the password also. This security measure will guarantee entry and subsequent stay of the people who are supposed to access the system and do business in the system.

The secure server will also consist of the user-friendly and secure interface where the user will be connected by the secure authentication system to the backend server. A proper authentication of the users transfers them to their dashboards depending on their positions in the system.

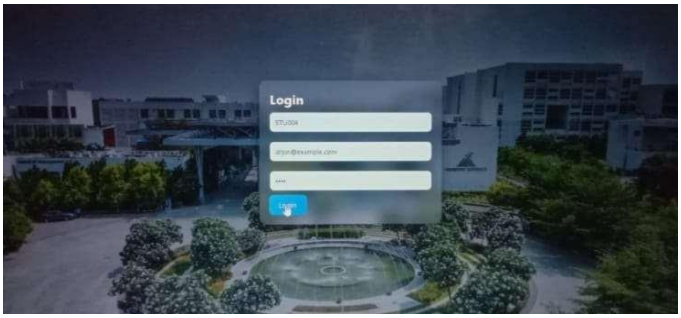

Figure 2: Advanced proposed Alumni Management system Log-in interface.

B. Student Direction, Right of Access to Information.

The interface center would be Student Dashboard where the students would be able to deliberate on the personal details, the upcoming activities and employment opportunities that would be announced by the alumni. The name, university ID, and email address and mentor status of a student will be added to the dashboard as all the significant information.

The system gives the students the option of reading the events of the institution and making their bookings through the platform. Second, the vacancies posted by the alumni professional and students might also become familiar with the current career opportunities and internships that are posted in the dashboard.

The specified characteristic will help enhance access to the information, related to the career, and will bring the former students and the current students closer.

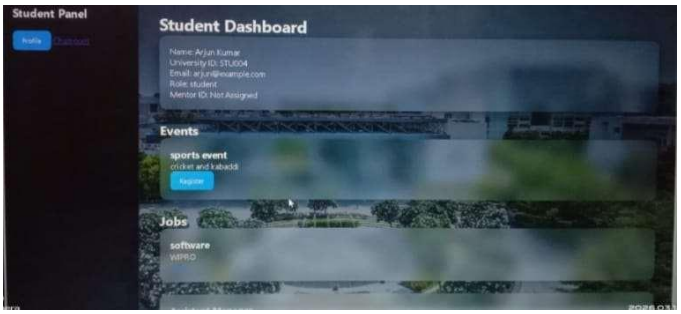

Figure 3: There is a display of Student Dashboard Profile, events, and job opportunities information.

C. interactive module of the alumni and mentors.

It also involves mentor management interface within which the professionals and the alumni can meet the students and answer their calls to be mentored. The module allows the mentors to see the profile of the students and give them career advice, among others, accessing the requests of the students.

The interface shall also show the number of students that a particular mentor has been assigned to and the mentor will have a choice of viewing the resume of a student. The system allows the alumni to post the employment opportunities that can be offered to the students in the job referral module.

This module improves the relationship between the alumni and information exchange between the current students and graduates.

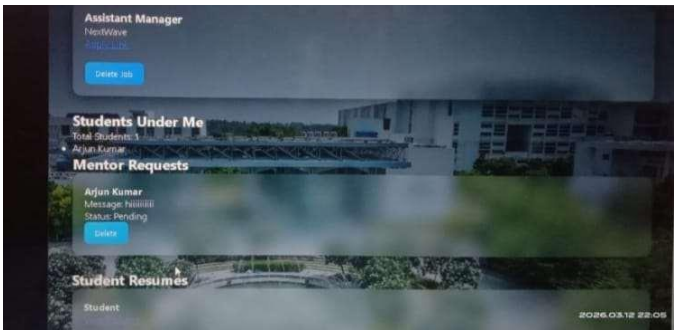

Figure 4: Student Requests/Mentorship Activities Alumni/Mentor panel.

D. System Performance and Effect.

These findings show that the suggested system is capable of incorporating a set of features in a centralized system. With the modular structure, communication between the data layer, presentation and application layer was operational and this caused the system to be operational.

The alumni database is central and this enhances the accuracy and availability of the information that gives the administrators an opportunity to monitor the engagements of the alumni. The system also connects the alumni to the students since they provide mentorship communication and job referral services.

In general, the new platform will improve relations management with the alumni, offer the development of the students careers, and have a structural approach towards interaction with the alumni.

VIII. CONCLUSION AND FUTURE WORK

This paper used the design and development of a Digital Platform to Centralized Alumni Data Management and Engagement to pursue the management of alumni information and improve the relationship between schools and colleges on one hand and their graduates on the other hand. The traditional ways of handling the alumni may be premised on the distributed databases and manual channels of communication that lead to lack of updated records and integration of the alumni. The proposed system will correct all these issues as it will offer a centralized digital application that will streamline the storage of alumni data, communication tools, mentorship engagement, and job referral services into a single organized system.

The platform is built on a multi-layered design that includes presentation, application, data, and analytics layers and which ensures scalability, security, and maintainability of the system. The outcomes of the implementation prove that the platform will result in the ability to manage the alumni data properly and connect the alumni with students. An environment of collaboration is also facilitated by allowing the student career development and alumni involvement through the student dashboard, mentorship requests, job referral modules, and administrative monitoring tools.

These results indicate that the proposed system can contribute to the enhancement of the institutional communication and growth of the access to career-related opportunities, which is facilitated by matching students and successful alumni professionals. Information in the central database will allow the precise and unified information and the modular system will allow the institutions to process the alumni records and activities of

interest without issue.

Generally, the system assists in the growth of stronger alumni networks, and institutional development by employing a methodical and data-driven approach to managing alumni.

Although the proposed system is a good solution when it comes to managing alumni, there are some additions that can be made to the system in future developments. The interaction of the machine learning and artificial intelligence technique might enable the formulation of the mentorship recommendations which are individualized, suitable to the job opportunities and prediction of the way the alumni will interact with others. The features would also make the platform better at career development and collaboration of the alumni.

As an alternative, one can also say that the elaboration of a mobile app can be introduced as the future task to make it more accessible and engage users. Data connectivity and functionality can also be improved by interconnection with the third party platforms such as professional networking applications, job portals and institutional student information systems. This would enable the platform to be more scalable and usable in assisting institutions to develop a more dynamic and sustainable system of alumni interaction.

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